



OFFICIAL

Midodrine for Postoperative Hypotension Guideline

1. Purpose

To provide guidance for the use of midodrine¹ oral vasopressor in supporting a patient's blood pressure in the first 24 hours post-operatively, to mitigate the vasodilatory effects of neuraxial or general anaesthesia.

2. Applicability

This guideline applies to all clinical staff working across surgical services at Armadale Kalamunda Group.

3. Background

Hypotension is a frequently occurring complication in the postoperative period and is associated with adverse outcomes^{2,5}. These include myocardial, renal and cerebral injury including delirium. Despite adequate resuscitation, many patients require short term blood pressure support to counteract the predictable hypotensive effects of anaesthesia.

In selected patients, midodrine may be initiated for blood pressure support in the perioperative period and continued for the first 24 hours only.

4. Guidelines

Midodrine may be initiated for postoperative hypotension under the supervision of a Consultant Anaesthetist in surgical patients ONLY. Other causes of hypotension such as hypovolaemia, sepsis, anaphylaxis, adrenal insufficiency and cardiogenic shock, MUST be excluded prior to commencement.

Treatment of postoperative hypotension is to be commenced in the Post-Anaesthetic Care Unit (PACU) or surgical ward within 24 hours post-operatively, by a Consultant Anaesthetist or nominated deputy anaesthetist.

Other causes of hypotension must be excluded and corrected in the first instance – most commonly hypovolaemia due to inadequate fluid resuscitation or haemorrhage³.

4.1 Indications – postoperatively

- Hypotension
- Presence of symptoms consistent with hypotension eg. Dizziness
- Anticipation of hypotension based on clinical assessment

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4.2 Contraindications

Absolute

- Active bleeding
- Hypovolaemia
- Sepsis
- Anaphylaxis
- Supine hypertension
- Impaired enteral absorption
- Pheochromocytoma
- Known hypersensitivity to any component of the product
- Acute kidney injury

Relative

- Impaired swallow reflex
- Severe congestive cardiac failure
- Severe renal impairment (eGFR <30)
- Urinary retention
- Narrow angle glaucoma
- Severe peripheral vascular disease
- Cerebrovascular occlusions and vessel spasms
- Proliferative diabetic retinopathy
- Thyrotoxicosis
- Long QT syndrome

4.3 Prescription requirements

- Midodrine must be prescribed on the WA Hospital Medication Chart (WAHMC)
- Initial dose of 10 to 20mg* to be prescribed in the “Once only” section and administered as a stat dose
- Further doses of 10mg* to be prescribed in the “Regular medicines” section at 4 hourly intervals following the stat loading dose, for five (5) further doses
- Indication is to be written as “post-op hypotension: do not give if systolic BP $\geq$ 130mmHg”
**
- Regular antihypertensive medications should be withheld while the patient is receiving midodrine
- Information regarding the commencement of midodrine to be handed over between PACU and ward staff, utilising the ISOBAR format as per AKG clinical handover procedure AK051/23

* Consider dose reduction if weight <50kg or renal impairment

** This systolic BP parameter may be altered up or down by the prescriber according to the patient’s usual pre-operative readings.

5. Monitoring requirements

Patients are to be monitored as per standard nursing guidelines using the Adult Observation and Response Chart AKMR140.3. Blood Pressure and Pulse Rate is to be checked immediately prior to the administration of midodrine and the prescribed dose omitted if systolic blood pressure is equal to or above the parameter indicated by the prescriber.

6. Supporting information

The following documents support the implementation of this policy:

- [Medication Management Procedure](#)
- [Clinical Handover Procedure](#)
- [Documentation Standards Procedure](#)

7. Compliance, monitoring and evaluation

The Head of Department Anaesthetics will be responsible for monitoring compliance and clinical incidents via the Datix CIMS pathway and audit of prescription, administration, and patient outcomes.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Medication Safety
- Standard 5 – Comprehensive Care
- Standard 6 – Communicating for Safety
- Standard 8 – Recognising and Responding to Acute Deterioration

9. References

1. WA Medicines Formulary, midodrine tablet
2. Matthew D McEvoy, Ruchir Gupta, Elena J Koepke, Aarne Feldheiser, Denny Levett, Julie K. M. Thacker, Mark Hamilton, Michael P. W. Grocott, Monty G Mythen, Timothy E. Miller and Mark R. Edwards for the POQI-3 workgroup. Perioperative Quality Initiative consensus statement on postoperative blood pressure, risk and outcomes for elective surgery. BJA 122 (5) 575-586 (2019)
3. Verna M Aykanat, Ian O Fleming, Tomás B Corcoran: Midodrine and its Potential Role in Postoperative Hypotension. Australian Anaesthesia (the Blue Book) [https://www.anzca.edu.au/resources/college-publications/australasian-anaesthesia-\(the-blue-book\)/blue-book-2019-\(1\)](https://www.anzca.edu.au/resources/college-publications/australasian-anaesthesia-(the-blue-book)/blue-book-2019-(1)) pp 101-110
4. Australian Government Department of Health Therapeutic Goods Administration, Prescription Medicines, regulatory decisions and notices: Midodrine
5. Sessler DI, Meyhoff CS, Zimmerman NM, Mao G, Leslie K, Vasquez SM, et al. Period-dependent associations between hypotension during and for four days after noncardiac surgery and a composite of myocardial infarction and death: a substudy of the POISE-2 trial. Anesthesiology. 2018;128(2):317-27.

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
Midodrine	An oral preparation licensed for the management of orthostatic hypotension ¹ . The pharmacodynamic action as a α -1 adrenoceptor agonist causes peripheral vasoconstriction ⁴ .
Peri-operative period	From holding bay admission up to 24 hours post-operatively
Hypotension	Systolic blood pressure $\geq$ 20% below the patient's baseline usual preoperative reading
Anaesthesia	Neuraxial, general or sedation in the context of surgery

11. Document contact

Head of Anaesthesia

Enquiries relating to this guideline may be directed to:

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12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V2	28 11 2025	28 11 2028	Full review, minor formatting only.

13. Authorisation

This guideline has been authorised and issued by the A/Director Clinical Services.

Authorised by	Dr John O'Hare - Deputy Director Clinical Services
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